

(32-bit)

Decimal sayıların doubleword - length 2'ye tımlanması

a) 3874

Binary: 0000 0000 0000 0000 0000 1111 0010 0010

Hex: 0 0 0 0 0 F 2 2

3874 0

1937 1

968 0

484 0

242 0 1111 0010 0010

121 1

60 0

30 0

15 1

7 1

3 1

1

b) 10000000

Binary: 0000 0000 0000 1111 0100 0010 0100 0000

Hex: 0 0 0 F 4 2 4 0

1000000 0

500000 0

250000 0

125000 0

62500 0

31250 0

15625 1

7812 0 1111 0100 0010 0100 0000

3906 0

1953 1

976 0

488 0

244 0

122 0

61 1

30 0

15 1

7 1

3 1

1

c) -100 → 100' on hex hali → 00000064

10 11 12 13 14 15
A B C D E F

↓
16' 6 + 16' 4

$$\begin{array}{r} \overset{\text{F F F F F F F } 16}{100000000} \\ - 00000064 \\ \hline \text{FFFFFF9C} \end{array}$$

Binary: 1111 1111 1111 1111 1111 1111 1001 1100

Hex: FFFFFFF9C

d) -5555 → 5555 hex hali

$$\begin{array}{l} 16^3 = 4096 \\ 16^2 = 256 \\ 16^1 = 16 \\ 16^0 = 1 \end{array}$$

$$\left. \begin{array}{l} 1 \times 4096 \\ 5 \times 256 \\ 11 \times 16 \\ 5 \times 1 \end{array} \right\} (15B3)_{16}$$

$$\begin{array}{r} \overset{\text{F F F F } 16}{100000000} \\ - 000015B3 \\ \hline \text{FFFFEA4D} \end{array}$$

Binary: 1111 1111 1111 1111 1110 1010 0100 1101

Hex: FFFFEA4D

e) -4 → 4 hex hali → 00000004

$$\begin{array}{r} \overset{\text{F F } 16}{100000000} \\ - 00000004 \\ \hline \text{FFFFFFFC} \end{array}$$

Binary: 1111 1111 1111 1111 1111 1111 1111 1100

Hex: FFFFFFFC

f) 32767

$$\begin{array}{r} 32767 \quad 1 \\ 16383 \quad 1 \\ 8191 \quad 1 \\ 4095 \quad 1 \\ 2047 \quad 1 \\ 1023 \quad 1 \\ 511 \quad 1 \\ 255 \quad 1 \\ 127 \quad 1 \\ 63 \quad 1 \\ 31 \quad 1 \\ 15 \quad 1 \\ 7 \quad 1 \\ 3 \quad 1 \\ 1 \end{array}$$

Binary: 0000 0000 0000 0000 0111 1111 1111 1111

Hex: 0 0 0 0 7 F F F

16-bit

Decimal sayıların byte-length 2'ye tımlanması

a) 23

Binary: 0000 0000 0001 0111

b) 111

 $64+32+8+4+2+1$

Binary: 0000 0000 0110 1111

c) -100 hex hali $\rightarrow$ 0064

$$\begin{array}{r} \overset{16}{10000} \\ - \overset{16}{0064} \\ \hline \text{FF9C} \end{array}$$

Binary: 1111 1111 1001 1100

d) -55 hex hali $\rightarrow$ 0037 $3 \cdot 16 + 7 \cdot 1$

$$\begin{array}{r} \overset{16}{10000} \\ - \overset{16}{0037} \\ \hline \text{FFC9} \end{array}$$

Binary: 1111 1111 1100 1001

e) 99

99	1
49	1
24	0
12	0
6	0
3	1
1	

Binary: 0000 0000 0110 0011

f) -15 hex hali $\rightarrow$ 000F

$$\begin{array}{r} \overset{16}{10000} \\ - \overset{16}{000F} \\ \hline \text{FFF1} \end{array}$$

Binary: 1111 1111 1111 0001

16-bit word-length sayılar hem 2'ye tünleyen sayı, hem de işaretli olabilir. Karşılaştığımız decimal bul.

Unsigned

Unsigned değer her sayının 10'luk karşılığıdır.

Signed

MSB 0 ise → sayı pozitif
MSB 1 ise → sayı negatif (2'ye tünleyen sayıdır)

Negatifler:

$$\text{Signed değer} = \text{Unsigned değer} - 65536$$

→ 2^{16} den

a) 00 A3 → 0000 0000 1010 0011

$$16^3 \cdot 0 + 16^2 \cdot 0 + 16^1 \cdot 10 + 16^0 \cdot 3 = 163$$

Unsigned : 163

MSB 's' 0 yani pozitif

Signed : +163

b) FF FE → 1111 1111 1111 1110

$$16^3 \cdot 15 + 16^2 \cdot 15 + 16^1 \cdot 15 + 16^0 \cdot 14 = 65534$$

MSB 's' 1 yani negatif

Unsigned : 65534

$$\rightarrow \text{Signed} = 65534 - 65536 = -2$$

Signed : -2

c) 6F 2D → 0110 1111 0010 0000

$$16^3 \cdot 6 + 16^2 \cdot 15 + 16^1 \cdot 2 + 16^0 \cdot 0 = 28448$$

Unsigned : 28448

MSB 's' 0 yani pozitif

Signed : +28448

d) B6 4A → 1011 0110 0100 1010

$$16^3 \cdot 11 + 16^2 \cdot 6 + 16^1 \cdot 4 + 16^0 \cdot 10 = 46666$$

MSB 's' 1 yani negatif

$$\text{Signed} = \text{Unsigned} - 65536$$

Unsigned : 46666

$$\text{Signed} = 46666 - 65536 = -18870$$

Signed : -18870

e) AB 05 $\rightarrow$ ^{MSB} 1010 1011 0000 0101

$$16^3 \cdot 10 + 16^2 \cdot 11 + 16^1 \cdot 0 + 16^0 \cdot 5 = 43781$$

MSB's 1, yes: negativ

$$\begin{aligned}\text{Signed} &= 43781 - 65536 \\ &= -21755\end{aligned}$$

Unsigned: 43781

Signed: -21755

f) 53 D2 $\rightarrow$ ^{MSB} 0101 0011 1101 0010

$$16^3 \cdot 5 + 16^2 \cdot 3 + 16^1 \cdot 13 + 16^0 \cdot 2 = 21458$$

MSB's 0, yes: positiv

Unsigned: 21458

Signed: +21458